

VQE convergence: FlagQuantum simulation vs. Quafu Baihua

$H = -Z_0Z_1 - X_0 - X_1$ • physical qubits Q123/Q124 • 1,024 shots/basis



SERIES

- FlagQuantum Statevector
- FlagQuantum NoiseModel (noise from Baihua)
- Exact ansatz minimum
- Quafu Baihua hardware (95% shot CI)

QUAFU BAIHUA CALIBRATION
Captured 2026-08-06 19:07:05

Q123 T1 36.414 μ s T2 68.655 μ s
1Q fidelity 0.999

Q124 T1 27.679 μ s T2 9.455 μ s
1Q fidelity 0.999

Q123-Q124 CZ fidelity 0.991
CZ duration 224 ns
1Q gate duration 64 ns

FlagQuantum NoiseModel $\rightarrow$ hardware
MAE 0.0446
RMSE 0.0534